

ZORIAN THORNTON

Ph.D Candidate, Genome Sciences, Seattle, WA
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EDUCATION

University of Washington - Seattle, WA Ph.D in Genome Sciences (Expected Winter 2025/2026)	<i>Sep. 2019 - Present</i>
Virginia Tech - Blacksburg, VA B.Sc Statistics	<i>Aug. 2015 - May 2019</i>
Virginia Tech - Blacksburg, VA B.Sc Computational Modeling and Data Analytics Minors in Mathematics and Computer Science	<i>Aug. 2015 - May 2019</i>

PROGRAMMING LANGUAGES, TOOLS, AND CONCEPTS

Machine Learning	Deep learning frameworks (PyTorch), transformers, supervised/unsupervised methods, bayesian inference, anomaly detection
Data Processing	Python, R, pandas, numpy, SQL, data pipelines, statistical analysis, reproducible research workflows
Software Engineering	Test-driven development (TDD), software architecture & refactoring, configuration management, type-safe systems design
Cloud & Deployment	Version control (Git), workflow management (Snakemake), CI/CD pipelines
HPC	SLURM, CUDA, openMP, performance optimization
Data Visualization	matplotlib, seaborn, ggplot2, interactive visualization techniques
Development Environment	Unix, bash, collaborative version control, AI-assisted development, code review & quality assurance
Other Skills	Java, MATLAB, dimensionality reduction techniques (PCA, MDS, UMAP, t-SNE)

RESEARCH AND WORK EXPERIENCE

Fred Hutchison Cancer Research Center Doctoral Candidate, advised by Frederick Matsen IV	<i>Seattle, WA</i> <i>July 2021 - Present</i>
• Developing end-to-end machine learning pipelines to predict viral fitness and evolution using large-scale biological data • Designed and implemented biophysically-inspired neural networks (PyTorch) that enable accurate predictions of protein variant phenotypes from experimental data • Applied statistical methods to analyze complex biological datasets and extract meaningful patterns • Collaborated with research groups to validate models and improve computational approaches	
Adaptive Biotechnologies Machine Learning Computational Biology Intern	<i>Seattle, WA</i> <i>June 2022 - Sept. 2022</i>
• Implemented and benchmarked semi-supervised learning methods for classification of T-cell receptor specificity using high-throughput sequencing data	

- Designed data preprocessing pipelines to handle complex biological sequence data at scale
- Conducted systematic downsampling experiments to optimize model performance across varying data conditions
- Developed visualization tools to communicate technical findings to cross-functional teams

Virginia Tech Department of Statistics

Undergraduate Research with Leah Johnson

Blacksburg, VA

Aug. 2017 - May 2019

- Developed predictive environmental models integrating temperature-dependent biological parameters with disease transmission dynamics to forecast Bluetongue virus spread across different climate scenarios
- Built Bayesian hierarchical models using MCMC methods to quantify uncertainty in environmental-disease relationships, incorporating temperature-sensitive vector life history traits and host-pathogen interactions

University of Washington Department of Genome Sciences

Undergraduate Research with Jim Bruce

Seattle, WA

June. 2017 - Aug. 2017

- Visualized inter-surface regions of cross-linked proteins and built probabilistic models to quantify competition between proteins for linkage with other proteins to help scientists characterize protein function, discover mutations, and discover protein-protein interactions to tackle molecular challenges such as cancer

PUBLICATIONS

- **Thornton, Z.**, Haddox, H.K., Tran, T, Bedford, T, Matsen IV, F.A. (2025). Simulating the genetic and antigenic evolution of viruses under selection pressure from host immunity *Manuscript in preparation*
- Yu, T., **Thornton, Z.**, Hannon, W., DeWitt, W.S., Radford, C., Matsen IV, F.A. and Bloom, J.D. (2022). A biophysical model of viral escape from polyclonal antibodies *Virus Evolution*, 8(2), veac110.
- El Moustaid, F., **Thornton, Z.**, Slamani, H., Ryan, S. and Johnson, LR (2021). Predicting temperature-dependent transmission suitability of bluetongue virus in livestock. *Parasites Vectors*, 14(1), pp.1-14..
- Keller, A., Chavez, J.D., Eng, J.K., **Thornton, Z.** and Bruce, J.E. (2018). Tools for 3D Interaction Visualization. *Journal of proteome research*, 18(2), pp.753-758.

RESEARCH PRESENTATIONS

- Zorian Thornton, *Predicting Disease Progression of Colorectal Cancer via Self-Organizing Maps*, RECOMB 2019, George Washington University, Washington DC, May 2019
- Zorian Thornton, *Modeling Bluetongue Virus via Markov Chain Monte Carlo Methods*, Student Experiential Learning Conference, Virginia Tech, Blacksburg, VA, Apr. 2018
- Zorian Thornton, *Viewing Molecular Interaction Interfaces Through Directed Computational Methods*, Department of Genome Sciences Research Symposium, University of Washington, Seattle, WA, Aug. 2017